

5

GEOGRAPHY

Population

Contemporary India - I
(NCERT Book)



Simply stated, population is a crucial component that quantifies the quantity of individuals (or organisms of a particular species) residing in a given location. It is a crucial piece of knowledge that must be understood before to creating and carrying out plans for a society's development.

It supports the research and development of resource distribution plans and programs by political scientists, health professionals, and other policy analyst. The necessity of measuring the population, tracking its growth patterns, and determining its density in order to design and implement developmental strategies is explained in the following chapter.

Topic Notes

- ☐ Significance of Population
- ☐ Population Size and Distribution
- ☐ Population Growth and Processes of Population Change

Population is a predominant element of a society. It is a reference point that defines the usage of resources in an area. It helps in the formulation of various developmental schemes and policies. Population provides meaning to the available resources and cultural elements in a region. The study of population stands on an estimate of its size, its distribution in a region and a change in its pattern. The following chapter talks about the importance of the estimation of the population. It also throws light upon the population size and density of India.

TOPIC 1

SIGNIFICANCE OF POPULATION

Learning Objectives

- Students will be introduced to concepts like population, resources, census, etc.
- Students will learn about population as an element of a society.
- Students will read about the significance of humans as producers and consumers of resources on Earth through a few examples.
- Students will read about three major parameters/questions associated with population— size, distribution, processes of population change and qualities of a population.

Learning Outcomes

- Students will be able to describe the significance of humans and their functions on Earth.
- Students will be able to understand the meaning and significance of population and census.
- Students will be able to enlist various parameters of the population.

Humans have utilised the available natural resources to create a social and cultural environment tailored to their whims and fancies. They exploit and use these resources and are themselves, a type of resource with varying qualities. Their contribution is essential to develop the economy and society.

Coal was just a piece of rock until people were able to invent technology to exploit it and make it a 'resource'. Natural events like floods or tsunamis are labelled as 'disasters' only when they affect a crowded village or town. Hence, population is pivotal element in social studies.

Example 1. Mention two ways in which humans are central to the definition of a resource.

(Understand)

Ans. Two ways in which humans are central to the definition of a resource are:

- (1) Humans have utilised these resources to create a comfortable and luxurious society that supports their development.
- (2) Resources like diamond and coal would only be rocks unless humans exploited them and used them for their benefit.

The numbers, distribution, growth, characteristics and qualities of the human population provide a background to understand and appreciate all aspects of the environment.

Human beings are producers and consumers of earth's resources. Thus, it is important to be aware of the number of people living in a region, their lifestyles and characteristics, an increase/decrease in their numbers, etc. The census provides us with that information.

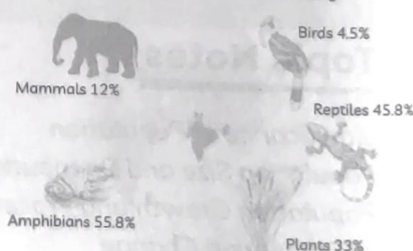


Frequently Asked

→ Three major parameters and factors associated with population are:

- (1) Population size and distribution
- (2) Population growth and processes of population change
- (3) Characteristics or qualities of the population— Age, sex, health, etc.

Endemism in India's Biodiversity



Important

→ Population is the point of reference from which all other elements are observed and from which they derive significance and meaning.



TOPIC 2

POPULATION SIZE AND DISTRIBUTION

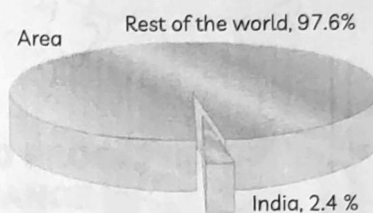
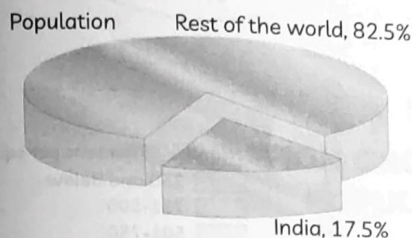
Learning Objectives

- Students will learn various facts about the size of the Indian population.
- Students will read about the patterns of population distribution—by number, density, etc.
- Students will read some statistics about the uneven distribution of population in India.

Learning Outcomes

- Students will be able to define the terms like population density, size of population, etc.
- Students will be able to understand the reasons behind the uneven distribution of population.

In March 2011, India's population was 1,210.6 million, which accounts for 17.5 per cent of the world's population. India's population is unevenly distributed over the country's vast area of 3.28 million square km (about 2.4 per cent of the total area of the world).



According to the 2011 census data, Uttar Pradesh is the most populous state of India with a population of 199 million. It houses about 16 per cent of the country's population. The Himalayan state of Sikkim has a population of just about 0.6 million and Lakshadweep has only 64,429 people.

Almost half of India's population lives in the five states of Uttar Pradesh, Maharashtra, Bihar, West Bengal and Andhra Pradesh.

Important

- Rajasthan, the biggest state in terms of area, has only 5.5 per cent of the total population of India.

India's Population Distribution by Density

Population density can be defined as the number of people per unit area. India is one of the most densely populated countries in the world.

Important

- Bangladesh and Japan have higher average population densities than India.

The population density of India in the year 2011 was 382 persons per sq km. In India, densities across states vary from 1,102 persons per sq km in Bihar to only 17 persons per sq km in Arunachal Pradesh.

Population densities differ because of the following reasons:

- Rugged terrain and unfavourable climatic conditions discourage people from living in a region. Hence, there is a sparse population in Uttarakhand and most of the north-eastern states of India.
- Assam and most of the Peninsular states have moderate population densities because of the hilly, dissected and rocky nature of the terrain. Other influences on the density of this region are moderate to low rainfall and the presence of shallow and less fertile soils.
- The Northern plains and Kerala in the south have high to very high population densities because of the flat plains with fertile soils and abundant rainfall.

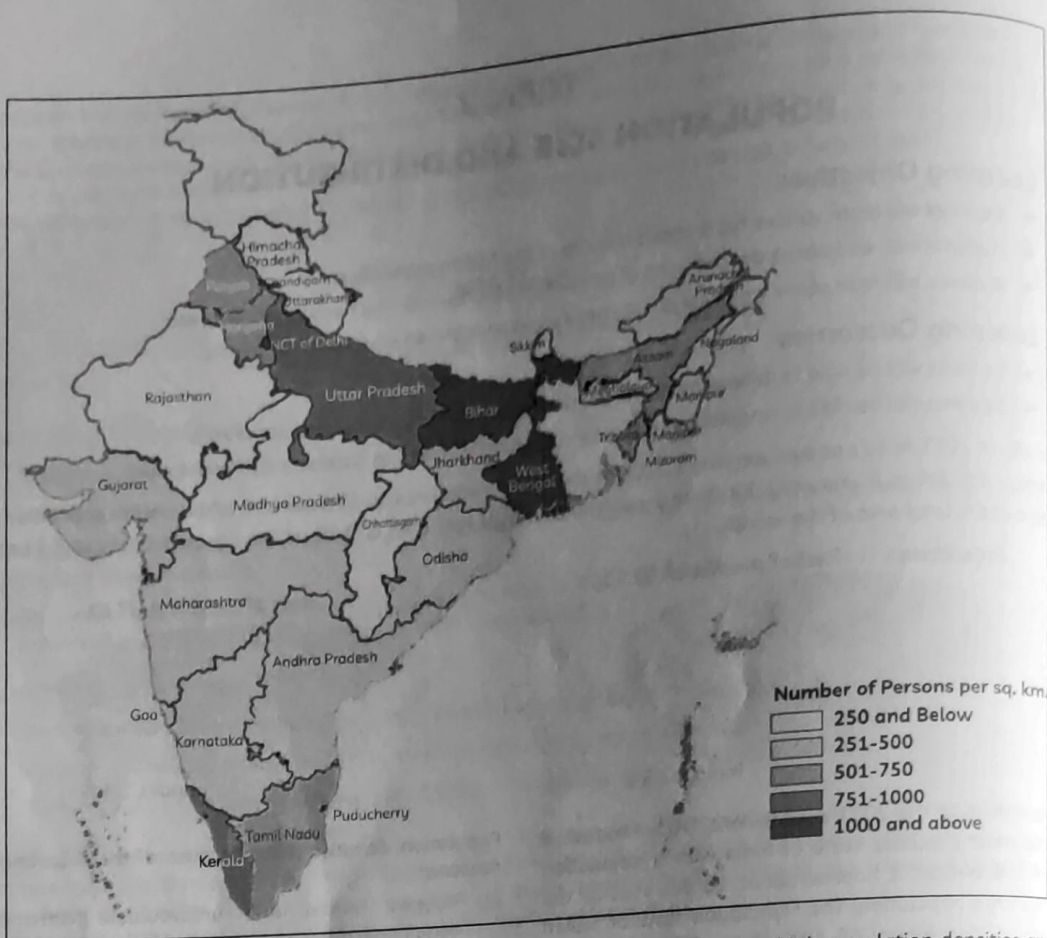
Example 2. What could be the reason of uneven distribution of population in India? (*Understand*)

[NCERT]

Ans. India has an uneven distribution of population across the states. There are several reasons which cause this uneven distribution, such as:

- Difference in birth rates and death rates
- Topography and rugged terrain, which lead to migration in certain areas.
- Climatic conditions



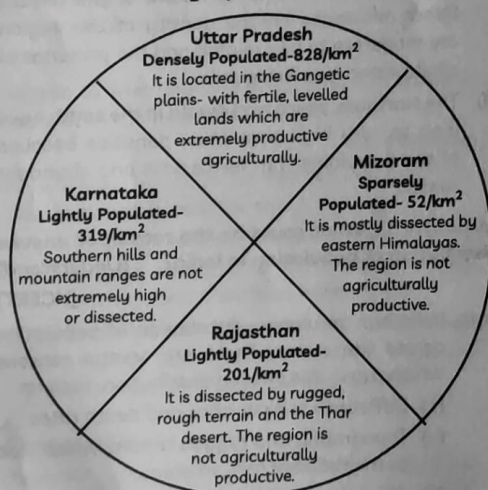


Example 3. Name three states located in the Northern Plains with high population densities.
(Remember)

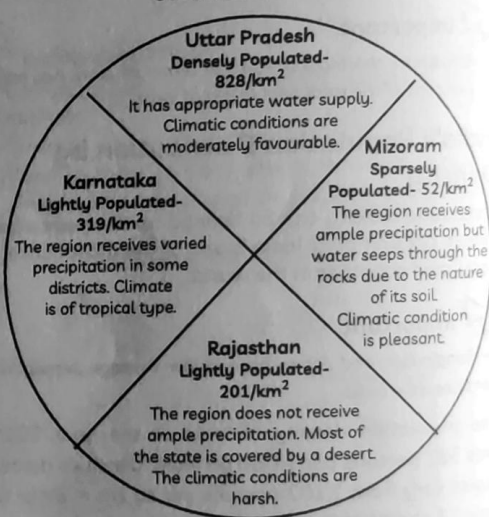
Ans. Three states with high population densities are Uttar Pradesh, Bihar and West Bengal.

Reasons behind the Uneven Distribution of Population across India

Geographical Factor

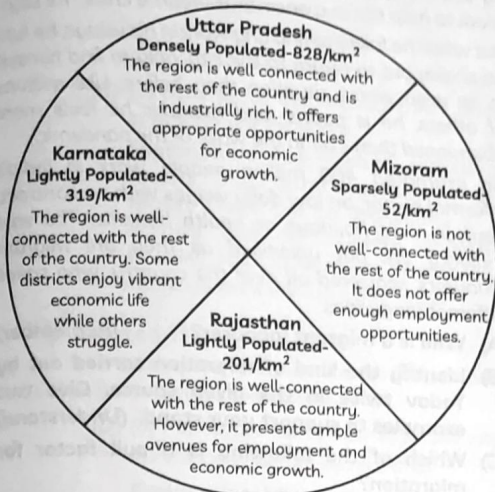


Climatic Controls

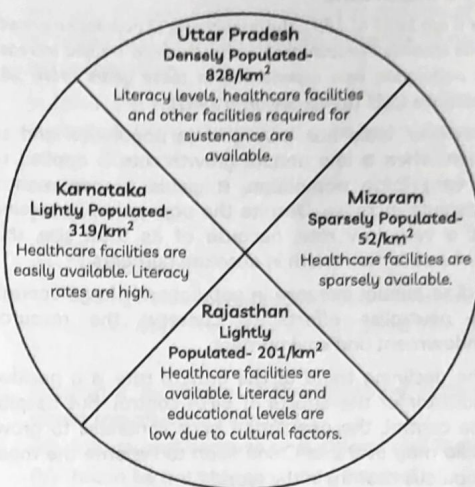


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Transportation Facilities



Facilities for Sustenance



TOPIC 3

POPULATION GROWTH AND PROCESSES OF POPULATION CHANGE

Learning Objectives

- Students will read about the growth in the population of a region— factors responsible for it.
- Students will learn about the three processes that contribute to population growth.
- Students will learn the formulas to measure population growth.

Learning Outcomes

- Students will be able to understand the concept of population growth.
- Students will be able to describe the three processes that influence population growth and their individual and collective influence upon the population of a country.
- Students will be able to calculate the annual and absolute growth rates with appropriate data.

Population growth is a dynamic function, which means that it is dependent on a number of factors and keeps changing constantly.

The interaction of the following three processes influences the growth of population:

- (1) Births
- (2) Deaths
- (3) Migration

Population Growth

Population growth can be experienced in a geographical location over time. The growth of population can be understood as the change in number of inhabitants of a country or territory over a period of time. This change can be indicated either in absolute numbers or percentage change per year.

The absolute number of people which is added each year is the magnitude of the increase.

It can be calculated by simply subtracting the previous population from the current population. It is referred to as the absolute increase.

The pace of population increase is calculated in per cent per annum. This can be understood as the annual growth rate. For example, if there was a rise of 10% in population in a region, it would mean that in a given year, 10 people were born or added to the population for every 100 people living in this place.

India's population has seen steady growth from 361 million in 1951 to 1210 million in 2011. However, since 1981, the rates of growth have started declining gradually as a result of declining birth rates.



Important

➔ From 1951 to 1981, the annual rate of population growth was steadily increasing; as a consequence, a rapid increase in population was experienced in these years (from 361 million in 1951 to 683 million in 1981).

However, India has a very large population and so even when a low annual growth rate is applied to a very large population, it yields a monumental absolute increase. Despite the population increasing at a very low rate, because of its total size, this increase is mammoth in absolute numbers.

India's annual increase in population is large enough to neutralise efforts to conserve the resource endowment and environment.

The declining trend of the growth rate is a positive indicator of the efforts of birth control. But despite the control, the population base continues to grow. India may overtake China soon to become the most populous country in the world.

Processes of Population Change/Growth

Three main processes that continue to influence the growth of the population can be understood as follows:



Frequently Asked

➔ The natural increase in population is the difference between birth rates and death rates.

Birth rate can be defined as the number of live births per thousand persons in a year. Birth rates are usually higher than death rates. It is a major component of the growth of the population.

Death rates can be defined as the number of deaths per thousand people in a year. Decline in death rates is usually the reason for increase in population.

When we consider the population growth trends of India till 1980, high birth rates and declining death rates, led to a spurt of population growth. Since 1981, birth rates have started declining gradually, bringing down the growth in the population with them.

Migration: Migration can be defined as the movement of people from one place to another for settlement. Migration is both internal (within the country) and international (between the countries). Internal migration only influences the distribution of population within the nation. It doesn't affect the size of the population.

Migration causes a significant change in the composition and distribution of population.

Example 4. Case Based:

Read the following passage and answer the questions that follow.

The day Yadav, 34, a construction worker, reached his village, he vowed never to return to the city. "I felt betrayed twice: by society, because no one around me lent a hand — my landlord kicked me out — and

by the state. I trusted the Prime Minister Narendra Modi to help me in a once-in-a-lifetime crisis," he says. But when he failed to find any work in his village, he had no choice but to return to the city, only to find himself in an even worse situation than before. Like millions of others, he is poorer and hungrier he feels more abandoned than ever in the wake of the pandemic.

An estimated 400 million people work in India's informal sector, on low daily wages with no contract, pension, paid holidays or health benefits. The vast majority are not unionised as they are migrant labourers, scattered all over the country, who speak different languages.

- (A) Who is a migrant labourer? (Remember)
- (B) Identify the kind of migration carried out by Yadav twice in the given source. Give two examples to support your stand. (Understand)
- (C) Which of the following is a pull factor for migration?
- Object poverty
 - Lack of resources in a place
 - Absence of good schools in proximity
 - Jobs up to the management level in a company are available (Understand)
- (D) How did Yadav's migration to his village affect the population composition of India?
- No change
 - Population size of India was reduced
 - Population composition of the city and his village changed permanently.
 - Population composition of the city and his village changed temporarily. (Apply)
- (E) Assertion (A): Like millions of others, Yadav is poorer, hungrier and feels more abandoned than ever.

Reason (R): This is because of the sudden population explosion in the cities.

- Both (A) and (R) are true and (R) is the correct explanation of (A).
- Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (A) is correct but (R) is wrong.
- (A) is wrong but (R) is correct. (Analyse)

Ans. (A) A migrant labourer is a casual and unskilled worker who migrates systematically from one region to another, offering their services on a temporary basis.



Caution

➔ Even though some terms mentioned in the sources are not taught or explained in the chapter, students must try to read the source multiple times to understand their meanings and formulate a definition in their own words.

- (B) The kind of migration carried out by Yadav in the given source is internal migration. This can be proved by the following statements:

- (1) Yadav moved from a city to his village and back within India. He didn't move to a foreign country.
- (2) Yadav's movement was temporary.
- (C) (d) Jobs up to the management level in a company are available

Explanation: Pull factors include those factors that attract people to a city/region. There can be availability of better schools and employment opportunities in cities. Here, availability of jobs upto the management level is a pull factor.



Related Theory

→ Other alternatives are push factors that encourage a person to move away from a place.

- (D) (d) Population composition of the city and his village changed temporarily.

Explanation: Migration causes a significant change in the composition and distribution of the population. Internal migration only affects the composition of a place temporarily. Yadav moved from the city to town and back temporarily.

- (E) (C) (A) is correct but (R) is wrong.

Explanation: There was no sudden population expansion in the cities. Population growth rates have been declining steadily. Yadav felt poorer because of sudden migration and increased competition for resources, progress and place to live.



Related Theory

→ Increased competition rises from rising population but it also rises from scarcity of resources and competition for growth.

In India, people usually migrate from rural to urban areas because of various "push" factors that make rural areas unattractive for residential or employment purposes. Some of these factors are:

- (1) Scarce opportunities for employment
- (2) Poverty
- (3) Limited resources and potential for growth
- (4) Limited exposure to the corporate world

The "Pull" Factors of the cities on the other hand, are:

- (1) Increased employment opportunities
- (2) Better living conditions
- (3) Opportunities for growth and development
- (4) Better connectivity
- (5) Better availability of educational and professional facilities

As a determinant of population, migration changes not only the population size but also the composition of urban and rural populations in terms of age and sex composition.

In India, the rural-urban migration caused a steady increase in the percentage of population in cities and towns. The urban population increased from 17.29 percent of the total population in 1951 to 31.80 percent in 2011.



Important

→ There has been a significant increase in the number of 'million plus cities' from 35 to 53 in 10 years, from 2001 to 2011.

Example 5. Which of the following statements about migration is incorrect?

- (a) It is caused because of push and pull factors.
- (b) It can be both external and internal.
- (c) It can only affect the size of a population.
- (d) Migration can be temporary or permanent.

(Analyse)

Ans. (c) It can only affect the size of a population.

Explanation: Migration can affect both the size and the distribution of a population. It can also change the density of population at a place because it often involves mass transfers of people from one place to another.



Caution

→ Students must pay attention to each word mentioned in the question. They must not get confused between correct and incorrect. Keywords should be underlined during the first reading of a question.

Adolescent Population

The size of the Indian adolescent population is by far its most notable characteristic. It makes about one-fifth of India's overall population. Generally speaking, adolescents fall between the ages of 10 and 19. They are the most important resource for the future. Adolescents have greater nutritional needs than the average child or adult. Growth retardation and deficiencies can result from inadequate nutrition.

However, the cuisine that teenagers in India can eat is deficient in every nutrient. A significant portion of teenage girls get anemia. Their issues have not gotten enough attention in the development process thus far. It is necessary to make the teenage girls more aware of the issues they face. The dissemination of education and literacy among them can raise awareness.

National Population Policy

In 1952, the Indian government launched a comprehensive family planning program after realising that family planning would enhance people's health and well-being. The goal of the Family Welfare Program has been to encourage planned and responsible parenthood on a voluntary basis. Years of deliberate work culminated in the National Population Policy (NPP) 2000.

The NPP 2000 lays out a framework for policies that include providing free and mandatory education up to the age of 14, lowering the infant mortality rate to less than 30 per 1000 live births, vaccinating all children against vaccine-preventable diseases, encouraging girls to delay marriage, and making family welfare a people-centered initiative.

Judicious use of Natural Resources and Suggests Measures for their Conservation

India's natural resources are under tremendous strain due to its large and expanding population. Sustainability is threatened by air pollution, deforestation, and excessive water consumption. People must embrace environmentally friendly practices like energy efficiency, waste segregation, and water harvesting in order to ensure conservation. Societies need to encourage less plastic use, sustainable agriculture, and afforestation. The government needs to invest in renewable energy, implement stringent environmental regulations, and promote citizen involvement. India can only strike a balance between resource protection and growth for future generations by taking collective responsibility.



Glossary

- (1) **Census:** A census is a survey conducted on the full set of observation objects belonging to a given population.
- (2) **Birth rate:** The number of live births per thousand of population per year is called birth rate.
- (3) **Death rate:** The number of deaths in a given period divided by the population exposed to the risk of death in that period.
- (4) **Migration:** Migration can be defined as the temporary movement of a population in search of better living conditions.
- (5) **Population density:** The number of individuals living within that specific location determines the population density, or the number of individuals divided by the size of the area.
- (6) **Annum:** A year.
- (7) **Terrain:** A stretch of land, especially with regard to its physical features.
- (8) **Demographic:** Relating to the structure of the population.
- (9) **Enumeration:** Counting/Numbering.
- (10) **Resource:** A stock or supply of money, materials, staff, and other assets that can be drawn on by a person or organisation in order to function effectively.
- (11) **Million plus cities:** Urban agglomeration with a population of one million or more.

OBJECTIVE Type Questions

[1 mark]

Multiple Choice Questions

1. Radha heard a news debate about the growing number of cars in Delhi which causes air and noise pollution. She comes home and asks her father whether people

in Delhi have started buying more cars than they did before. Her father replies that the number of cars in Delhi is rising because of the increase in population. Radha asks her father the reason behind this growth.

Help Radha's father answer her question. Which of the following are plausible reasons behind the growth of the population in Delhi?

- (I) High birth rates
- (II) Low death rates
- (III) Better climatic conditions
- (IV) Better employment opportunities
- (V) Easy connectivity to nearby metro cities

- Options:
- (a) (I), (II) and (III)
- (b) (II), (III) and (IV)
- (c) (I), (II), (IV) and (V)
- (d) (I) and (V)

Ans. (c) (I), (II), (IV) and (V)

Explanation: The primary reasons for the rise in population in any place are increasing birth rates, declining death rates and migration. Migration is possible when the location possesses pull factors, one of which is better employment opportunities. Delhi offers better employment opportunities and people from nearby cities migrate to Delhi, leaving their rural regions behind.

Related Theory

- Delhi is facing severe air pollution, which indicates that its climatic conditions are not better than its neighbouring areas.
- Migration is caused by various push and pull factors.

2. Which is the least populated state of India?

- (a) Tripura
- (b) Sikkim
- (c) Kerala
- (d) Himachal Pradesh [Delhi Gov. QB 2024]

Ans. (b) Sikkim

Explanation: Sikkim has the lowest population among the states of India. Sikkim houses only 0.6 million people. Among the union territories of India, Lakshadweep has the lowest population with only about 64,429 people.

Related Theory

- Uttar Pradesh has the highest population in the country.

3. Every ten years, the Government of India collects detailed information about the country's population, including age, gender, occupation, and literacy rate. What is this official process called?

- (a) Population density
- (b) Birth rate
- (c) Census
- (d) Population rate

Ans. (c) Census

Explanation: Census can be defined as the official enumeration of the population done periodically. Census of a place can define its demographic structure which can help with better policy planning.

Related Theory

- The first census of India was conducted in 1872. However, the first complete census was taken in the year 1881.

4. Which of the following countries is most likely to have a greater population density than India?

- (a) China
- (b) Pakistan
- (c) Sri Lanka
- (d) Bangladesh

Ans. (d) Bangladesh

Explanation: India has an extremely high population density. However, Bangladesh and Japan have a higher population density than India. They are smaller than India in size.

Related Theory

- Population density is the number of people living in a place per unit area.

5. Which of the following is a positive indicator of birth control in a population?

- (a) Declining birth rate
- (b) Declining death rate
- (c) Migration due to pull factors
- (d) Rising population density

Ans. (a) Declining birth rate

Explanation: Declining birth rate is the only factor which ensures birth control or growth control of the population. The rest of them are not related to control in birth notes.

6. Read the data given below and answer the question:

Rate of Population Growth of India			
Year	Total Population (in million)	Absolute Increase in the Decade (in million)	Annual Growth Rate(%)
1981	683.3	135.17	2.22
1991	846.4	163.09	2.16
2001	1028.7	182.32	1.97
2011	1210.6	181.46	1.64

From the table, what inference can be deducted for Annual Growth Rate of population?

- (a) Increasing
- (b) No change
- (c) Decreasing
- (d) None of the above [Delhi Gov. QB 2024]

Ans. (c) Decreasing

Explanation: The annual growth rate of 2.22 in 1981 has been decreased to 1.64 in 2011.

7. Which of the following states has a very high population density?

- (a) West Bengal (b) Gujarat
- (c) Telangana (d) Arunachal Pradesh

Ans. (a) West Bengal

Explanation: West Bengal has one of the highest population densities in India. Most of the land in West Bengal is covered with fertile alluvial soil brought down by the river Ganga and its numerous tributaries. This soil is highly suitable for agriculture and hence attracts population.

8. Migrations change the number, distribution and composition of the population in:

- (a) The area of departure
- (b) The area of arrival
- (c) Both the area of departure and arrival
- (d) None of the above [DIKSHA]

Ans. (c) Both the area of departure and arrival

Explanation: Migration involves the movement of population. This change affects both the region of arrival as well as departure. In migration, the overall population remains the same. Migration just changes the population composition of a place.



Related Theory

Urban areas have seen a boom in population, since they provide better standards of living within certain parameters. This increase in population has resulted in high population density.

9. The areas having are always highly populated.

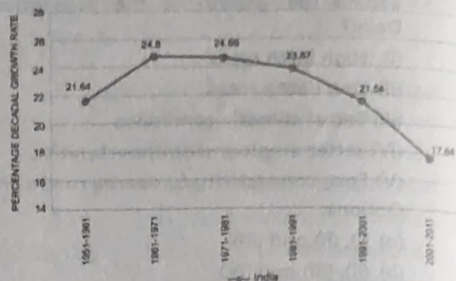
- (a) rugged terrains
- (b) fertile soils and flat plains
- (c) hilly and rocky terrain
- (d) less fertile and low rainfall areas

[DIKSHA]

Ans. (b) fertile soils and flat plains

Explanation: For a sustainable life, fertile soil and flat plains are priorities because they produce enough food and provide opportunities for development. One of the earliest civilisations was earliest established on the floodplains of the Indus and Gangetic plains. Hence, the areas with fertile soils and terrains usually shelter high populations.

10. Observe the given picture and choose the odd one out.



India's Population Growth Rates during 1951-2011

- (a) 1951-1961 (b) 1981-1991
- (c) 1961-1971 (d) 2001-2011

Ans. (c) 1961-1971

Explanation: Based on this given picture, the 1961-1971 was the last decade to see an exponential spurt of population growth. Thereafter, population growth rates have been steadily declining. This is why this decade is the odd one out.



Related Theory

The population is still growing, despite the decline in growth rate. The decline in growth rate means that the growth has slowed down and not that the growth of population has stopped.



Caution

It is important for the students to apply different logics and concepts to figure out the odd one from the list of alternatives.

11. Which of the following statements is correct?

- (I) Human beings are producers and consumers of the earth's resources.
- (II) The census of India provides us with information regarding the female population in India.
- (III) The first census was held in 1882.
- (IV) The Indian Census is the most comprehensive source of demographic, social and economic data.

Options:

- (a) (I), (II) and (III) (b) (II) and (III)
- (c) (I) and (IV) (d) (I) and (II)

Ans. (c) (I) and (IV)

Explanation: The census of India provides us with information regarding the demographic, social and economic data of the population. The first census was held in 1872. The first complete census was held in 1881.





Related Theory

Humans are the point of reference from which all other elements are observed and from which they derive significance and meaning. 'Resources', 'calamities' and 'disasters' are all meaningful only in relation to human beings.

12. Which of the following statements is an incorrect statement regarding population?
- Population rises every year.
 - It is restricted to the number of one species living in a place.
 - Population is a static phenomenon.
 - Population growth rates are declining due to birth control.

Ans. (c) Population is a static phenomenon.

Explanation: Population is a dynamic phenomenon. The numbers, distribution and composition of a population are constantly changing due to the influence of the interaction of the three processes, namely — births, deaths and migrations.

13. When a low annual rate is applied to a very large population, it yields a large absolute increase. Which of the following statements is true regarding the given statement?
- Even though the annual growth rate is low, the population is still rising.
 - Even though the population growth rate is low, the number of people getting added to the population every year is still very high.
 - The population is not increasing.
 - Even though the annual growth rate is low, the population is declining.

Ans. (b) Even though the population growth rate is low, the number of people getting added to the population every year is still very high.

Explanation: A small percentage of a huge number is still a considerably high number. Thus, even though the growth rates are low, a considerable number of people are still added to the population every year.



Caution

Students must pay special attention to every keyword mentioned in the question. They must understand each concept clearly and apply them.

14. Consider the statements given below and choose the correct answer:

Statement (I): Sikkim is least populated state in India.

Statement (II): Lakshadweep is least populated UT in India.

- Statement (I) is correct and (II) is incorrect.
- Statement (I) is incorrect and (II) is correct.
- Both (I) and (II) are incorrect.
- Both (I) and (II) are correct.

Ans. (d) Both (I) and (II) are correct.

Explanation: Lakshadweep is the least populated UT in India with only 64,429 people living in the region.



Caution

Students must not mistake Lakshadweep to be a state. It is not a state. It is a union territory of India.

15. The magnitude of population growth refers to which of the following measures?

- The number of persons added to a region without any migration
- The total population of an area
- The rate of population growth
- The number of females per thousand males

Ans. (b) The total population of an area

16. Which of the following states is most likely to have the highest population density?

- UP
- Mizoram
- Karnataka
- Rajasthan

Ans. (a) UP

Explanation: Out of the given alternatives, UP is most likely to have the highest population density. Its current population density is 828 km². It is located at a plain, levelled area which is agriculturally productive.



Related Theory

Other states are not as densely populated because they are bordering states and suffer through other infirmities (difficult climatic conditions, less availability of water etc).

17. Which of the following statements about population are correct?

- It measures the number of people in an area.
- It changes over time.
- It is affected by climatic conditions, availability of educational and medical facilities.
- Rugged terrain witness dense population.

Options:

- (I) and (II)
- (II) and (III)
- (I), (II) and (III)
- (II) and (IV)

Ans. (c) (I), (II) and (III)

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Related Theory

India's population as of March 2011 stood at 1,210.6 million, which accounts for 17.5 percent of the world's population.

18. is the number of deaths per thousand persons in a year.

- (a) Death Rate (b) Migration
(c) Census (d) Absolute Rate

Ans. (a) Death rate

Explanation: The death rate counts the number of people who die every year in a particular place. It decreases the population in a country.

19. Census in India is held regularly every

- (a) 15 years (b) 5 years
(c) 10 years (d) 11 years

Ans. (c) 10 years

Explanation: A census is held every ten years. The last census was held in 2011.

20. Observe the given image and identify one of its benefit.



- (a) Increase in job opportunities
(b) Increase in birth rates
(c) Increase in immunisation of children
(d) Promoted delayed girl marriage

Ans. (c) Increase in immunisation of children

Explanation: Immunisation programs help protect children from life-threatening diseases such as polio, measles, and hepatitis. By ensuring timely vaccinations, these programs reduce infant mortality rates and improve overall public health.



Related Theory

The NPP 2000 lays out a framework for policies that include providing free and mandatory education up to the age of 14, lowering the infant mortality rate to less than 30 per 1000 live births, vaccinating all children against vaccine-preventable diseases, encouraging girls to delay marriage, and making family welfare a people-centered initiative.

21. Arrange the states according to their population distribution in descending order.

- (I) West Bengal (II) Andhra Pradesh
(III) Bihar (IV) Gujarat

Options:

- (a) (I), (II), (III), (IV)
(b) (II), (I), (III), (IV)
(c) (IV), (III), (II), (I)
(d) (III), (I), (II), (IV)

Ans. (d) (III), (I), (II), (IV)

Explanation: (1) Bihar- 9%
(2) West Bengal-8%
(3) Andhra Pradesh-7%
(4) Gujarat-5%

22. Match the items in Column A to those of Column B.

Column A	Column B
(A) Bihar	(I) 16% of India's population
(B) Uttar Pradesh	(II) Densely populated
(C) Karnataka	(III) Sparsely populated
(D) Lakshadweep	(IV) 5% of India's Population

Options:

- (a) A-(III), B-(I), C-(IV), D-(II)
(b) A-(II), B-(I), C-(IV), D-(III)
(c) A-(II), B-(IV), C-(I), D-(III)
(d) A-(III), B-(IV), C-(II), D-(I)

Ans. (b) A-(II), B-(I), C-(IV), D-(III)

Assertion-Reason (A-R)

In the following questions, two statements in the form of an Assertion (A) and a Reason (R) have been put forward. Read both statements carefully and choose the most appropriate option:

- (a) Both (A) and (R) are true and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(c) (A) is correct but (R) is wrong.
(d) (A) is wrong but (R) is correct.

23. Assertion (A): Population growth can be defined as the increase in population per unit area.

Reason: (R) Birth rate, death rate and migration are the factors which influence population growth.



Ans. (d) (A) is wrong but (R) is correct.

Explanation: Population density is defined as the increase in population per unit area, while population growth is just the increase in population of a geographical location at a particular time.

24. Assertion (A): Population of a region stays static.

Reason (R): Climatic conditions can influence its distribution.

Ans. (d) (A) is wrong but (R) is correct.

Explanation: The population is a dynamic factor in an economy that keeps changing every moment, hence the assertion statement is wrong. However, climatic conditions of the given region can influence its distribution. So, the reason is correct.

25. Assertion (A): The main cause of rise in the Indian population has been the rapid declines in death rates.

Reason (R): Death rate is the number of deaths per thousand people in 10 years.

Ans. (c) (A) is correct but (R) is wrong.

Explanation: Rapid declines in death rate causes the Indian population to grow in numbers. The reason does not define death rates correctly. Death rate is the number of deaths per thousand people in a year.

26. Assertion (A): India's population as of March 2011 stood at 1,210.6 million.

Reason (R): India is the third largest country in the world in terms of population. [DIKSHA]

Ans. (c) (A) is correct but (R) is wrong.

Explanation: India is the second largest country in the world in terms of population. It may surpass China to become the largest country in the world in terms of population in coming years.

27. Assertion (A): People move to urban centres and cities from towns in large numbers every year.

Reason (R): Smart cities offer good employment opportunities and standard of living.

Ans. (a) Both (A) and (R) are true and (R) is the correct explanation of (A).



Related Theory

→ The push factors of rural regions and towns are poverty, lack of work opportunities, unemployment and underdevelopment, poor economic condition, lack of opportunities, exhaustion of natural resources and natural calamities, scarcity of cultivated land, inequitable land distribution, low agricultural productivity, etc.

CASE BASED Questions (CBQs)

[4 marks]

Read the following passages and answer the questions that follow.

28. Rugged terrain and unfavourable climatic conditions are primarily responsible for sparse population in these areas. Which states have density below 250 persons per square km? Assam and most of the Peninsular states have moderate population densities. Hilly, dissected and rocky nature of the terrain, moderate to low rainfall, shallow and less fertile soils have influenced population densities in these areas. The Northern plains and Kerala in the south have high to very high population densities because of the flat plains with fertile soils and abundant rainfall.

(A) Which state in southern India is least densely populated and why?

(B) How is population density dependent upon the climatic and geographical conditions of a region?

Ans. (A) Tamil Nadu in southern India is less densely populated because of these reasons:

- (1) It is highly urbanised. It is doing extremely well on family planning and birth control.
 - (2) It has a large area.
 - (3) It has various hilly and forested regions.
- (Any two points)

(B) Population density is directly dependent upon the climatic and geographical conditions of a region:

- (1) If a region has pleasant climatic conditions, less humidity and gentle winds, it will be densely populated. Pleasant weather conditions cause the population density of a place to rise.

- (2) Population density is also affected by geographical features. Plain geography invites more settlers. Undulating topography causes people to migrate to plain regions.

29. India's population as of March 2011 stood at 1,210.6 million, which accounts for 17.5 percent of the world's population. These 1.21 billion people are unevenly distributed over our country's vast area of 3.28 million square km, which accounts for 2.4 percent of the world's area. The 2011 Census data reveal that Uttar Pradesh with a population size of 199 million is the most populous state of India. Uttar Pradesh accounts for about 16 percent of the country's population. On the other hand, the Himalayan state of Sikkim has a population of just about 0.6 million and Lakshadweep has only 64,429 people.

(A) Why are people unevenly distributed in India?

- (a) Because of India's rocky terrain.
- (b) India's unique location.
- (c) India is a peninsula, so people are scared of living close to coasts.
- (d) India has various geographical features and diverse climatic conditions that make some regions very favourable for living while others are difficult for survival.

(B) Which state in India has the least population living in it and why?

- (a) Himachal Pradesh, because of the mountains.
- (b) Sikkim, because of the cold weather.
- (c) Lakshadweep, because it is an island.
- (d) Jharkhand, because of the forests.

(C) What makes Uttar Pradesh a favourable place to live?

- (a) It produces a lot of food.
- (b) It has well drained soils and plain geography.

(c) It has many roads.

(d) It is well connected to the rest of the country.

(D) Why is India densely populated?

(a) It houses 18% of the world's population on only 2.4% of the total area in the world.

(b) It is a tropical country.

(c) It is located in Asia.

(d) It has various geographical features and diverse cultures.

Ans. (A) (d) India has various geographical features and diverse climatic conditions that make some regions very favourable for living while others are difficult for survival.

Explanation: India's varied geographical features and climatic conditions make some regions very attractive with appropriate residential and economic opportunities.

(B) (b) Sikkim, because of the cold weather.

(C) (b) It has well drained soils and plain geography.

Explanation: The northern plains have well-drained soils and support appropriate agricultural activity. Building houses and factories is also easier in the northern plains. Other reasons are also applicable but this alternative is the most well-rounded option.

(D) (a) It houses 18% of the world's population in only 2.4% of the total area in the world.

Explanation: Other alternatives do not give appropriate reasons for India's high population density. India's population is higher than what its area can shelter or sustain.



Related Theory

India's population as of March 2011 stood at 1,210.6 million, which accounts for 17.5 percent of the world's population.

SHORT ANSWER Type-I Questions (SA-I)

[2 marks]

30. Population density varies over space and time. Which are the three most densely populated states in India? Mention their reason.

Ans. Uttar Pradesh, West Bengal, and Bihar are three states that shelter half of India's population. Suitability of soil type for crop production, accessibility of water resources are some of the reasons for dense population in these regions.





Related Theory

Some other reasons can be poor sanitation facilities, poor infrastructure and lack of public health, etc.

- 31. Sara's class teacher asked her to count the number of students in her class along with the details of their residences and their phone numbers. Give a name to this exercise and its benefit.**

Ans. Sara's class teacher has asked her to take a census of her class. Census is simply the count of a population. One benefit of the census taking exercise in India is:

It is a comprehensive source of social and economic data about the population of the country. It supplies a lot of information about the lifestyles of Indian citizens.

- 32. Who are the producers and consumers of Earth's resources? [Delhi Gov. QB 2022]**

Ans. Humans are the producers and consumers of Earth's resources. It is because of their exploitation and usage that resources are called by that name. However, other living organisms also play a role in maintaining the balance of natural resources through their interactions with the environment.

- 33. 'Migration is the factor that plays a significant role in changing the composition and distribution of the population.' Mention one reason for push migration.**

[Mod. Delhi Gov. QB 2022]

Ans. Migration changes the composition and distribution of the population in a specific region. Migration adds and subtracts people from a region. Push migration can occur because of poor employment opportunities offered by a place.



Related Theory

Some other reasons can be poor sanitation facilities, poor infrastructure and lack of public health, etc.

- 34. 'Jammu and Kashmir has low population density than Uttar Pradesh.' Give reasons.**

[Delhi Gov. QB 2024]

Ans. This is because of the following reasons present in Jammu and Kashmir:

- (1) They have rugged terrain which is not that suitable for living than plains.
- (2) Climate there was too cold which do not support livelihood there.
- (3) Due to rugged terrain transport infrastructure is not that well developed.
- (4) Agriculture is more supported in the plains.

- 35. What is the annual growth rate of the population?**

Ans. The annual average rate of change of population size, for a given country, territory or geographic area, during a year is the annual growth rate of the population.

- 36. 'Raj moved to Mumbai from his village because he found a better job.' Justify the statement by highlighting some other factors that might have encouraged him to shift to Mumbai.**

Ans. Raj moved to Mumbai because of a push factor—the availability of a better employment opportunities. Some other factors for his migration can be:

- (1) Availability of better schools for his children.
- (2) Bad sanitation and medical facilities in his village.

- 37. How has the rural-urban migration resulted urbanisation in India? Explain.**

[Delhi Gov. QB 2024]

Ans. Urbanisation is the movement of rural people in urban areas. People do this because:

- (1) Poverty and unemployment in rural areas.
- (2) Better living conditions, healthcare facilities, education and sanitation are available there.

- 38. Migration changes population size and the population composition of urban and rural populations. Justify the statement.**

Ans. The following is the impact of migration on a population:

- (1) It adds/subtracts people from a population.
- (2) It changes the population composition of both the regions—the place from which people are migrating and the place to which they are migrating.

- 39. Population is the number of people living in the region at a particular time. Trace the population trends in India after 1981.**

Ans. After 1981, the following statements trace the trends in population in India:

- (1) India has seen a steady growth of population since 1981.
- (2) The annual growth rate of the population of India has decreased despite a steady, absolute increase in the numbers.

- 40. Define the annual growth rate of a population.**



Ans. The annual growth rate of a population can be described as:

- (1) The annual average rate of change of population size, for a given country, territory, or geographic area, during a specified period is the annual growth of a population.
- (2) It is expressed as the ratio between the annual increase in the population size and the total population for that year, usually multiplied by 100.

41. © How do climatic conditions affect the population density of a region?

Ans. Climatic conditions affect the population density of a region because:

- (1) It determines the agricultural productivity of the region. Ample rainfall, incidence of sun rays etc all impact the agricultural productivity of the region.
- (2) Climatic conditions also affect the transportation of the region. Roadways are usually found in dissected terrains instead of railway lines.

Both of these conditions affect the population density of a region heavily.

SHORT ANSWER Type-II Questions (SA-II)

[3 marks]

42. 'The study of the population of a place is a pivotal element.' Substantiate the statement with illustrations.

Ans. The study of population is beneficial for society in the following ways:

- (1) Population is a point of reference through which public policy is drafted. All the planning of policies and infrastructure is based on the population structure of an area.
- (2) The number, distribution, growth and characteristics of a population provide a basic background for understanding all the aspects of the environment.
- (3) The population density measures the stress level on the resources. Measurement of population helps with equitable redistribution of resources.

43. Differentiate between the terms—
Population growth and Population change.

Ans.

	Population Growth	Population Change
(1)	Population growth refers to the change in the number of inhabitants of a country and its territory during a specific period of time.	Population change refers to the change in the number of people during a specific period of time. It can be positive or negative.
(2)	It is a measure of the increase in population.	It deals with any change, in population.

(3)	Population growth does not alter population composition.	Population change measures the alteration in population growth itself.
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44. A local government body conducts a census in a village every ten years to gather data on the number of people living in it. This helps in planning resources, infrastructure, and development projects.

Based on this, explain the basic features of a population.

Ans. The basic features of a population are:

- (1) Population density: The number of people living
- (2) Population size: number of people living in a region over a given period of time is called the size of the population.
- (3) Population is distributed in various patterns across different regions in a country. Some regions have an appropriate number of citizens while others are not populated enough.
- (4) Birth rate and death rate
- (5) Migration: Migration of people from one place to another affect the size and composition of a population.

(Any three points)

45. What does growth in population mean? Describe two ways in which it is calculated.

Ans. The growth of the population refers to the change in the number of inhabitants of a country during a specific period of time.

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It can be calculated and expressed in two ways:

(1) **In Absolute Number:** It is obtained by simply subtracting the previous population from the current population. The difference is the net number of individuals added to the population in the given time. It is referred to as the 'absolute increase'.

(2) **Annual Growth Rate:** When the rate or growth of population is studied in percent per annum, this is referred to as the annual growth rate.

46. 'People are an important resource.' Identify three reasons to prove the statement.

Ans. People are an important resource because:

- (1) They exploit natural resources and use them to create a comfortable, luxurious, technologically advanced and culturally rich society.
- (2) They develop the economy and society in various ways.
- (3) They are a point of reference from which all other elements in a society are observed and defined.

47. A group of students is planning a research trip to explore regions in India with the least human interference. While studying a population density map, they notice that some states have significantly fewer people per square kilometer.

Based on this, identify the three least densely populated states. Justify their population density by giving a reason.

Ans. The three least densely populated states in India are:

- (1) Sikkim
- (2) Arunachal Pradesh
- (3) Himachal Pradesh

These states are sparsely populated because of the following reasons:

- (1) Undulating topography and difficult terrains
- (2) Difficult climatic conditions
- (3) Underdeveloped connectivity between places

48. Define migration term used for population composition? Write its features with reference to India.

Ans. Migration is one of the major components of population growth. It is defined as movement of people from one place to another.

Migration is of two kinds:

- (1) **Internal migration:** Migration within the country or a specific region is known as internal migration. For example, children living in rural regions often move to towns in search of better educational facilities with their parents.
- (2) **International migration:** Migration of people to different countries is called international migration. For example, when people migrate to other countries in search of better employment opportunities, it is called an international migration.

LONG ANSWER Type Questions (LA)

[5 marks]

49. The Census of 2011 carried some important information about the changing pattern of distribution of population in India. Highlight some facts about the distribution of population in India according to Census 2011.

Ans. According to the Census 2011 data:

- (1) Uttar Pradesh with a population size of 199 million people is the most densely populated state in India.

- (2) The Himalayan state of Sikkim shelters only about 0.6 million making it the least populous state in the country.
- (3) Lakshadweep has a population of only 64,429 people.
- (4) About half of India's population lives in five states located in the Northern plains—Uttar Pradesh, Maharashtra, Bihar, West Bengal and Andhra Pradesh.

- (5) Rajasthan is the biggest state in terms of area but only houses 5.5 percent of the total population of India.

50. 'Population is pivotal to the study of social science.' Justify the statement.

Ans. Population is the pivotal element in social studies. This can be described in the following points:

- (1) It is the point of reference from which other elements in a society derive significance and meaning.
- (2) 'Resources', 'calamities' and 'disasters' are all meaningful only in relation to human beings.
- (3) Human beings are producers and consumers of the earth's resources. Thus, knowledge about their numbers and their distribution is extremely important.
- (4) They exploit resources and drive the development of a society.

- (5) Study and estimation of population is essential for policy making, resource redistribution and other analysis.

51. © 'Karnataka is a favourable state to live in.' Analyse the statement with valid points.

Ans. Karnataka has a population density of 319 persons per km. It is a favourable state to live in because:

- (1) It is located on the extreme southern end of India and has moderate climatic conditions.
- (2) Its terrain is accessible and not dissected by valleys and rivers.
- (3) The region is well-connected with the rest of the country. Some districts enjoy vibrant economic life and offer ample employment opportunities.
- (4) Healthcare facilities are easily available.
- (5) Literacy rates are high and, housing and sanitation facilities are easily available.

LONG ANSWER TYPE QUESTIONS (LA)

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Multiple Choice Questions

- 61 families in a small village in Rajasthan, Nathdwara, moved to Jaipur in 1991 to look for better opportunities. Which of the following is the most likely correct statement about this information?
 - Only the population size of Nathdwara changed.
 - Only the population composition of Nathdwara and Jaipur changed, their sizes remained the same.
 - There was no change in the population composition of Rajasthan.
 - Only the population composition of Jaipur changed. (Evaluate) 1
- In India, a census was held in 1991. According to its rules, when will the next census be held in India?
 - 2011
 - 2005
 - 1996
 - 2001
 (Understand) 1
- is the most densely populated union territory in India.
 - Andaman Islands
 - Delhi
 - Lakshadweep
 - Chandigarh
 (Evaluate) 1

4. Read the given source.

Anurag in a news article read that Arunachal Pradesh has a very low population density of only 17 persons per sq km, compared to Bihar's 1,102 persons per sq km. he wondered Why is the population density in Arunachal Pradesh so low?

- Because it is a desert.
 - Because it is a bordering state and people are afraid of China.
 - Because it has rugged topography and difficult climatic conditions.
 - Because it is a north-eastern state.
- (Understand) 1

Assertion-Reason (A-R)

In the following questions, two statements in the form of an Assertion (A) and a Reason (R) have been put forward. Read both statements carefully and choose the most appropriate option:

- Both (A) and (R) are true and (R) is the correct explanation of (A).
- Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (A) is correct but (R) is wrong.
- (A) is wrong but (R) is correct.

5. Assertion (A): Uttar Pradesh is the largest state in India.

Reason (R): It shelters 199 million people which is the largest in the country. (Analyse) 1

6. Assertion (A): A census is an official enumeration of population done periodically.

Reason (R): India's first census was carried out in 1971.

(Analyse) 1

Case Based Question

- Read the following passage and answer the questions that follow.

In India, most migrations have been from rural to urban areas because of the "push" factor in rural areas. These are the adverse conditions of poverty and unemployment in the rural areas and the "pull" of the city in terms of increased employment opportunities and better living conditions. Migration is an important determinant of population change. It changes not only the population size but also the population composition of urban and rural populations in terms of age and sex composition. In India, the rural-urban migration has resulted in a steady increase in the percentage of population in cities and towns.

- Mention one implication of migration on the population density of urban regions. (Understand) 1
- Mention the pull factors of an urban centre like Indore. (Understand) 1
- What is a million plus city? How does migration play a role in the formation of a million plus city? (Understand) 2

Short Answer Type-I Questions

8. How is the population measured? Write down the formula for measuring the absolute growth of the population in a region. (Understand) 2
9. Why is the population in India increasing despite the decline in its population growth rate? (Analyse) 2

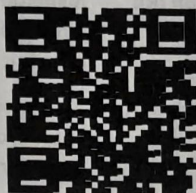
Short Answer Type-II Questions

10. Describe the three basic factors or processes that influence the population of a country. (Understand) 3

11. Differentiate between the total population and population density of a region using three points. (Analyse) 3

Long Answer Type Question

12. Ravi notices that while India's large population provides a strong workforce, it also leads to challenges like unemployment and resource depletion. In your opinion, how does a population of a country is important for development? (Understand) 5



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